# Geometric Sequences and Series

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## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** [Foundational] • Target: Geometric sequences

Consider the sequence $3, 6, 12, 24, \ldots$

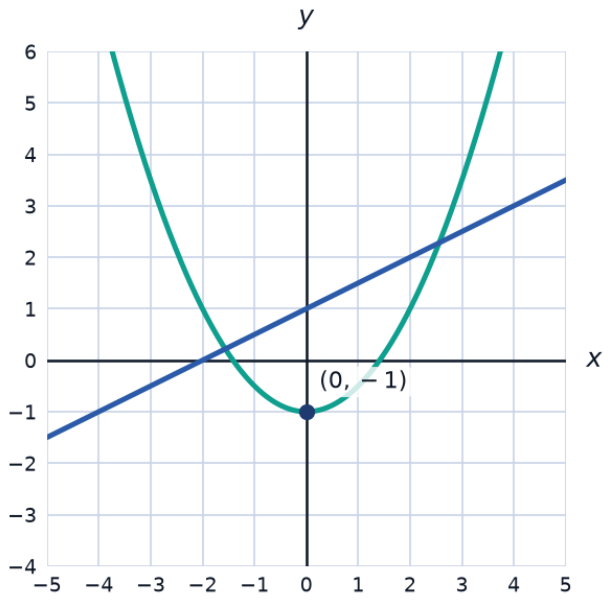
**a** State the common ratio.

**b** Write a formula for the $n$th term.

**c** Find $a_{10}$.

**2** [Foundational] • Target: Geometric sequences

In a geometric sequence, $a_2 = 6$ and $a_5 = 48$. Find the common ratio and the first term.




**3** [Foundational] • Target: Geometric series

Find the sum of the first 8 terms of $5 + 10 + 20 + \cdots$

**4** [Foundational] • Target: Geometric series

Evaluate each infinite sum, or state that it diverges:

**a** $18 + 6 + 2 + \frac{2}{3} + \cdots$

**b** $1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \cdots$

**c** $2 + 3 + 4.5 + \cdots$

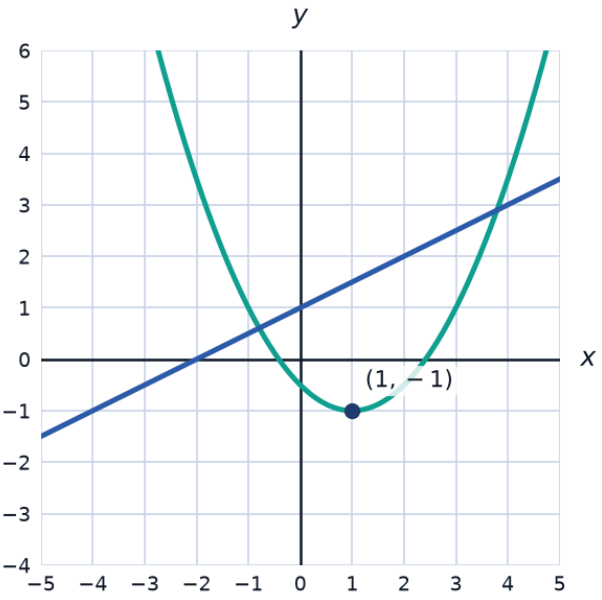
**5** [Foundational] • Target: Geometric series

Write the repeating decimal $0.\overline{7}$ as a fraction using an infinite geometric series.

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**6** [Foundational] • Target: Geometric series

A ball is dropped from 8 m and rebounds to $\frac{3}{4}$ of its previous height on each bounce. Find the total vertical distance it travels (falling and rising) before coming to rest.




**7** [Foundational] • Target: Geometric means and mixed sequences

Insert two positive geometric means between 4 and 108 (i.e., find $b_1, b_2$ so that $4, b_1, b_2, 108$ is geometric).

## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

**8** [Intermediate] • Target: Geometric means and mixed sequences

Determine whether each sequence is arithmetic, geometric, or neither:

**a** $2, 6, 18, 54, \ldots$ **b** $5, 8, 11, 14, \ldots$ **c** $1, 4, 9, 16, \ldots$

**9** [Intermediate] • Target: Applications

A car worth 80,000 riyals depreciates by 15% per year.

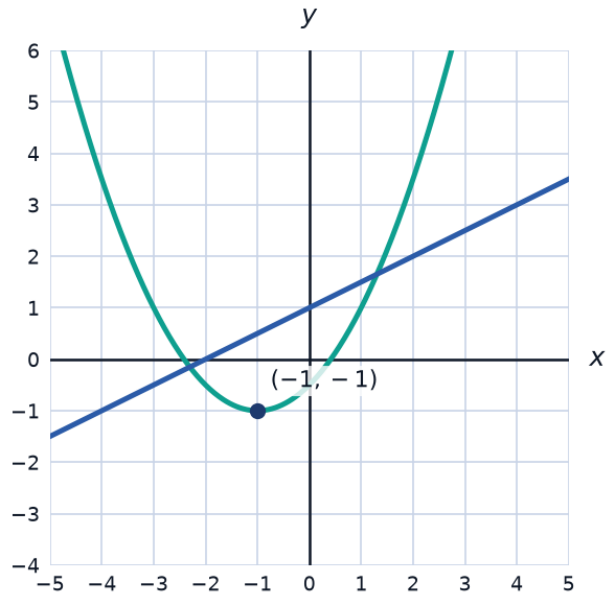
**a** Write a formula for its value after $n$ years.

**b** Find its value after 5 years, to the nearest riyal.

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**10** **[Intermediate]** • **Target: Applications**

A chain letter starts with 1 person sending it to 5 people, each of whom sends it to 5 more, and so on. Find the total number of people who have received the letter after 6 rounds (including the original sender's round).




**11** **[Intermediate]** • **Target: Visualizing geometric growth**

The first 6 terms of $a_n = 2(1.5)^{n-1}$ are plotted below. Contrast the shape of this graph with the straight-line shape of an arithmetic sequence's graph.

**12** **[Intermediate]** • **Target: Determining convergence before summing**

For each infinite geometric series, determine whether it converges WITHOUT computing the sum, by checking $|r| < 1$: $\sum_{n=1}^{\infty} 5(0.9)^{n-1}$, $\sum_{n=1}^{\infty} 3(1.2)^{n-1}$, $\sum_{n=1}^{\infty} 7(-0.4)^{n-1}$.

**13** **[Intermediate]** • **Target: Finding the number of terms from a partial sum**

For the geometric series $2 + 6 + 18 + \cdots$, find the number of terms $n$ needed for the sum to first exceed $2000$.

**14** **[Intermediate]** • **Target: A mixed application: population with geometric decline**

A population of an endangered species declines by 12% each year from an initial 5000. Find the total number of 'animal-years' lived by the population over the first 10 years (i.e. the sum of the population sizes at the start of each of the 10 years), to the nearest whole number.

## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

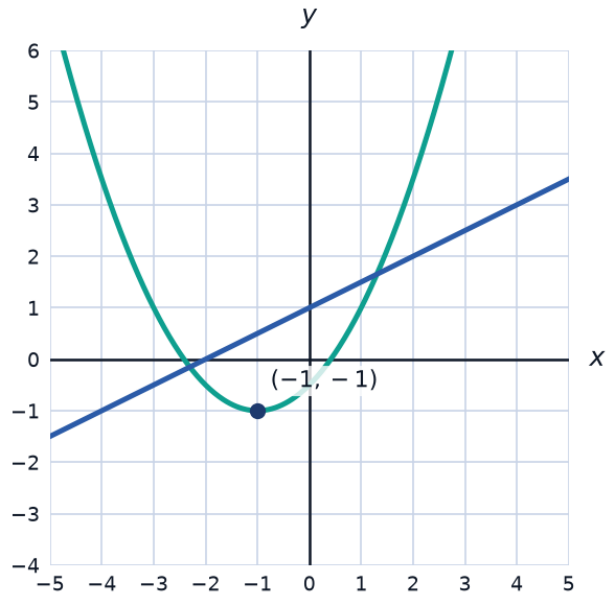
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**15** [Challenging] • Target: Geometric means and mixed sequences • Cross-Lesson: 4.03 Hyperbolas

Find the sum of the infinite geometric series with $a_1 = 12$ and $r = \frac{1}{3}$.

**16** [Challenging] • Target: Applications • Cross-Lesson: 5.01 Vectors: Components and Operations

Find the sum of the infinite geometric series with $a_1 = 20$ and $r = \frac{3}{4}$.




**17** [Challenging] • Target: Visualizing geometric growth • Cross-Lesson: 5.02 The Dot Product

Find the sum of the infinite geometric series with $a_1 = 18$ and $r = \frac{2}{3}$.

**18** [Challenging] • Target: Determining convergence before summing • Cross-Lesson: 5.03 Parametric Equations

Find the sum of the infinite geometric series with $a_1 = 8$ and $r = \frac{1}{2}$.

**19** [Challenging] • Target: Finding the number of terms from a partial sum • Cross-Lesson: 6.01 Polar Coordinates and Conversions

Find the sum of the infinite geometric series with $a_1 = 12$ and $r = -\frac{1}{3}$.

**20** [Challenging] • Target: A mixed application: population with geometric decline • Cross-Lesson: 6.02 Polar Equations and Graphs

Find the sum of the infinite geometric series with $a_1 = 6$ and $r = \frac{1}{4}$.

## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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**21** [Foundational] • Target: General Term of a Geometric Sequence

State the formula for the $n$-th term $a_n$ of a geometric sequence with first term $a_1$ and common ratio $r$.

**a.** $a_n = a_1 r^{n-1}$

**b.** $a_n = a_1 r^n$

**c.** $a_n = a_1 + (n-1)r$

**d.** $a_n = \frac{a_1}{r^{n-1}}$

**22** [Foundational] • Target: General Term of a Geometric Sequence

Find the 6th term of the geometric sequence $3, 6, 12, 24, \ldots$

**a.** 96

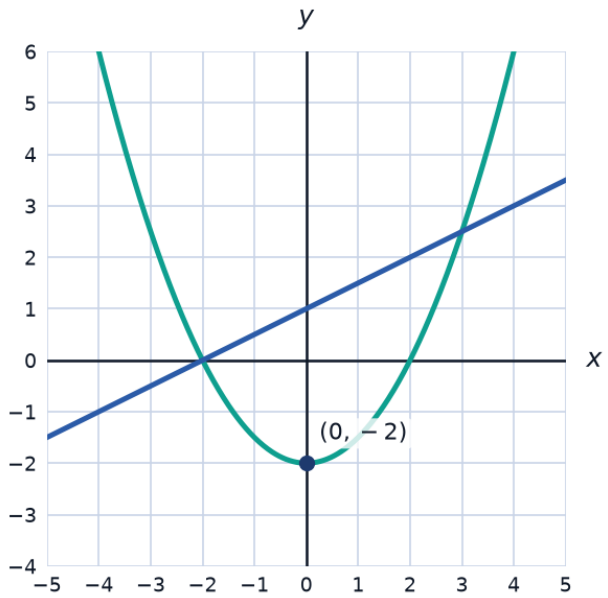
**b.** 192

**c.** 48

**d.** 128

**23** [Foundational] • Target: General Term of a Geometric Sequence

In a geometric sequence, $a_2 = 12$ and $a_5 = 324$. Find the common ratio $r$.




**a.** 3

**b.** 2

**c.** 4

**d.** $\sqrt{3}$

**24** [Foundational] • Target: Finite Geometric Series

State the formula for the sum $S_n$ of the first $n$ terms of a geometric series ($r \neq 1$).

**a.** $S_n = \frac{a_1(1-r^n)}{1-r}$

**b.** $S_n = \frac{a_1}{1-r}$

**c.** $S_n = \frac{n}{2}(a_1 + a_n)$

**d.** $S_n = a_1(1-r^n)$

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**25** [Foundational] • Target: Finite Geometric Series

Find the sum of the first 6 terms of the series $4 + 12 + 36 + \ldots$

**a.** 1456

**b.** 1458

**c.** 486

**d.** 2916

**26** [Foundational] • Target: Finite Geometric Series

Evaluate $\sum_{k=1}^{5} 2(3)^{k-1}$.

**a.** 242

**b.** 162

**c.** 243

**d.** 486

**27** [Foundational] • Target: Infinite Geometric Series

Under what condition does an infinite geometric series $\sum_{n=1}^{\infty} a_1 r^{n-1}$ converge?

**a.** $|r| < 1$

**b.** $r > 1$

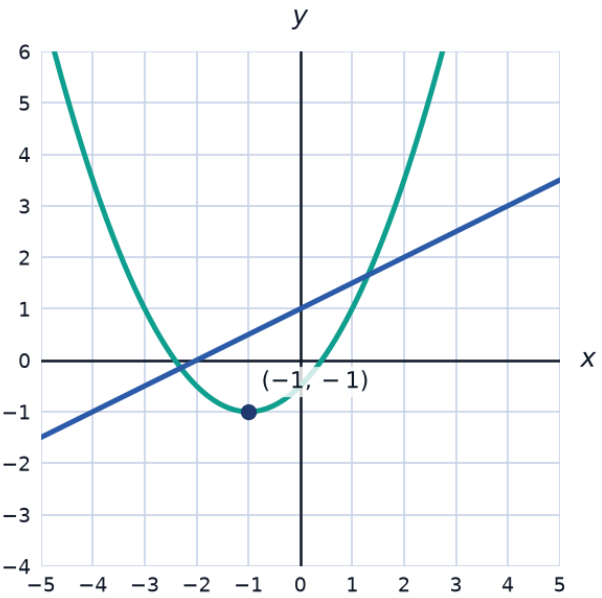
**c.** $|r| \leq 1$

**d.** $r > 0$

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**28** [Foundational] • Target: Infinite Geometric Series

Find the sum of the infinite series $12 + 4 + \frac{4}{3} + \frac{4}{9} + \ldots$




**a.** 18

**b.** 16

**c.** 24

**d.** 15

## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

**29** [Intermediate] • Target: Infinite Geometric Series

Find the fractional equivalent of the repeating decimal $0.\overline{45} = 0.454545\ldots$ using an infinite geometric series.

**a.** 5/11

**b.** 45/100

**c.** 9/22

**d.** 45/90

**30** [Intermediate] • Target: Geometric Word Problems and Annuities

A bouncing ball is dropped from a height of $16$ ft. On each bounce, it rebounds to $3/4$ of its previous height. Find the total vertical distance the ball travels before coming to rest.

**a.** 112 ft

**b.** 64 ft

**c.** 128 ft

**d.** 96 ft

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**31** [Intermediate] • Target: Geometric Word Problems and Annuities

A company offers a starting salary of $50000$ dollars with a guaranteed $4\%$ annual raise. What is the salary in the 5th year?

**a.** $50000(1.04)^4 \approx 58492.93$ dollars

**b.** $60000$ dollars

**c.** $58000$ dollars

**d.** $50000(1.04)^5$

**32** [Intermediate] • Target: Geometric Word Problems and Annuities

An ordinary annuity pays $100$ dollars at the end of each year for $10$ years at $5\%$ annual interest. Which expression represents its future value?

**a.** $100\left[\frac{(1.05)^{10}-1}{0.05}\right]$

**b.** $100(1.05)^{10}$

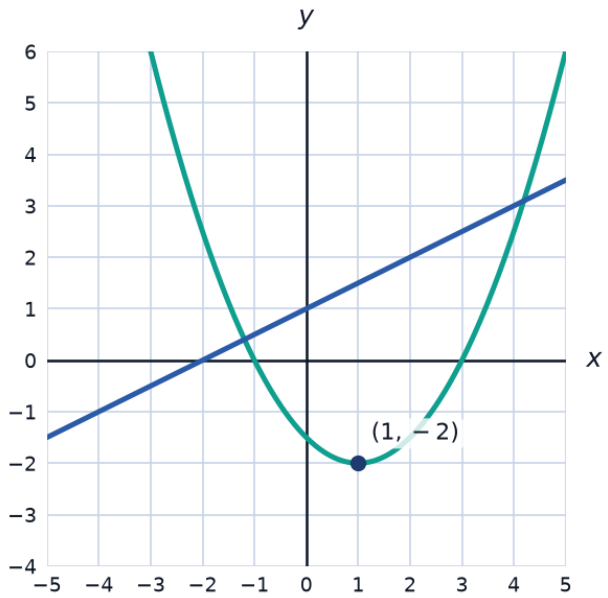
**c.** $1000(1.05)$

**d.** $\frac{100}{0.05}$

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**33** **[Intermediate]** • **Target: Geometric sequences**

Find the sum of the infinite geometric series with $a_1 = 4$ and $r = \frac{1}{2}$.




**a.** 10

**b.** 4

**c.** 8

**d.** 11

**34** **[Intermediate]** • **Target: Geometric series**

Find the sum of the first 5 terms of the geometric series with $a_1 = 1$ and $r = \frac{1}{2}$.

**a.** $\frac{15}{16}$

**b.** 5

**c.** $\frac{47}{16}$

**d.** $\frac{31}{16}$

**35** **[Intermediate]** • **Target: Geometric means and mixed sequences**

Find the sum of the first 3 terms of the geometric series with $a_1 = 3$ and $r = \frac{1}{2}$.

**a.** $\frac{17}{4}$

**b.** $\frac{25}{4}$

**c.** 9

**d.** $\frac{21}{4}$

**36** **[Intermediate]** • **Target: Applications**

Find the sum of the infinite geometric series with $a_1 = 5$ and $r = -\frac{1}{3}$.

**a.** $\frac{15}{4}$

**b.** $\frac{11}{4}$

**c.** 10

**d.** $\frac{19}{4}$

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**37** [Intermediate] • Target: Visualizing geometric growth

Find the sum of the infinite geometric series with $a_1 = 9$ and $r = -\frac{1}{3}$.

**a.** $\frac{23}{4}$

**b.** $\frac{27}{4}$

**c.** 18

**d.** $\frac{31}{4}$

## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

**38** [Challenging] • Target: Determining convergence before summing • Cross-Lesson: 5.03 Parametric Equations

Find the sum of the infinite geometric series with $a_1 = 8$ and $r = -\frac{1}{2}$.

**a.** $\frac{19}{3}$

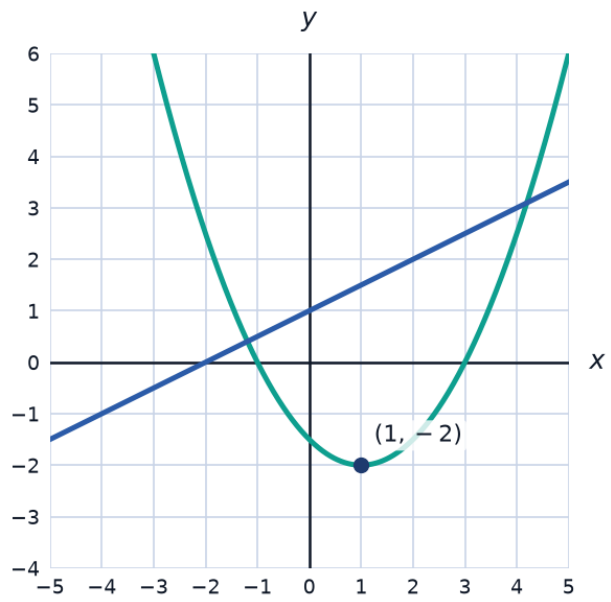
**b.** $\frac{13}{3}$

**c.** 16

**d.** $\frac{16}{3}$

**39** [Challenging] • Target: Finding the number of terms from a partial sum • Cross-Lesson: 6.01 Polar Coordinates and Conversions

Find the sum of the infinite geometric series with $a_1 = 6$ and $r = -\frac{1}{3}$.




**a.** $\frac{11}{2}$

**b.** $\frac{7}{2}$

**c.** 12

**d.** $\frac{9}{2}$

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**40** [Challenging] • Target: A mixed application: population with geometric decline • Cross-Lesson: 6.02 Polar Equations and Graphs

Find the sum of the infinite geometric series with $a_1 = 8$ and $r = \frac{1}{4}$.

**a.** $\frac{29}{3}$

**b.** $\frac{32}{3}$

**c.** $\frac{35}{3}$

**d.** 16

**41** [Challenging] • Target: General Term of a Geometric Sequence • Cross-Lesson: 7.01 Exponential Functions and Their Graphs

Find the sum of the infinite geometric series with $a_1 = 3$ and $r = -\frac{1}{3}$.

**a.** $\frac{9}{4}$

**b.** 6

**c.** $\frac{13}{4}$

**d.** $\frac{5}{4}$

**42** [Challenging] • Target: Finite Geometric Series • Cross-Lesson: 7.02 Logarithms and Their Properties

Find the sum of the infinite geometric series with $a_1 = 5$ and $r = \frac{1}{4}$.

**a.** 10

**b.** $\frac{23}{3}$

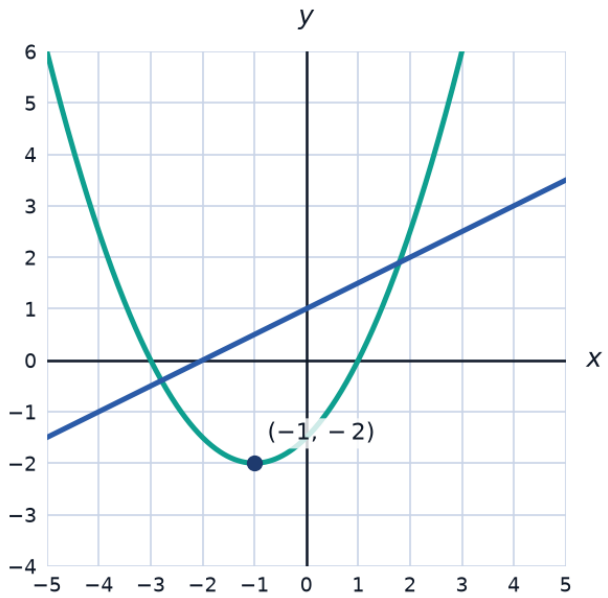
**c.** $\frac{17}{3}$

**d.** $\frac{20}{3}$

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**43** [Challenging] • Target: Infinite Geometric Series • Cross-Lesson: 7.03 Exponential and Logarithmic Equations

Find the sum of the infinite geometric series with $a_1 = 4$ and $r = -\frac{1}{2}$.




**a.** $\frac{5}{3}$

**b.** 8

**c.** $\frac{11}{3}$

**d.** $\frac{8}{3}$

**44** [Challenging] • Target: Geometric Word Problems and Annuities • Cross-Lesson: 7.04 Applications: Growth and Decay

Find the sum of the infinite geometric series with $a_1 = 2$ and $r = -\frac{1}{3}$.

**a.** $\frac{1}{2}$

**b.** $\frac{5}{2}$

**c.** 4

**d.** $\frac{3}{2}$

**45** [Challenging] • Target: Geometric sequences • Cross-Lesson: 8.01 Rational Functions and Asymptotes

Find the sum of the infinite geometric series with $a_1 = 3$ and $r = \frac{1}{3}$.

**a.** $\frac{11}{2}$

**b.** $\frac{9}{2}$

**c.** 6

**d.** $\frac{7}{2}$

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